

INTERNSHIP PROJECT REPORT

California Housing Price Prediction Using Linear Regression

A complete end-to-end supervised machine-learning pipeline built on the California Housing dataset, covering exploratory data analysis, OLS regression, evaluation metrics, and a model improvement roadmap.

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Program	B.Tech — CSE (Artificial Intelligence & Machine Learning)
Institution	ACE Engineering College
Dataset	California Housing (sklearn / Pace & Barry, 1997)
Model	Ordinary Least Squares — LinearRegression
Report Date	May 2026

R² = 0.7565

Variance Explained

MAE = \$52,670

Mean Absolute Error

RMSE = \$65,820

Root Mean Sq Error

Abstract

This report presents a supervised machine-learning pipeline that predicts California block-level median house values using eight socioeconomic and geographic features. After thorough exploratory data analysis (EDA), a StandardScaler–LinearRegression pipeline was trained on **16,512 samples** and evaluated on **4,128 held-out samples**, achieving **$R^2 = 0.7565$** , **MAE = \$52,670**, and **RMSE = \$65,820**. Key findings include the dominant predictive power of median income (MedInc), the coastal geographic premium reflected in coordinate coefficients, and the ceiling effect of the \$500k data cap on model performance. A structured improvement roadmap — covering polynomial features, regularised models, tree-based ensembles, and geospatial clustering — is presented to guide future iterations toward an R^2 target of 0.85+.

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1. Exploratory Data Analysis (EDA)

1.1 Dataset Overview

The California Housing dataset (Pace & Barry, 1997) contains **20,640 records** derived from the 1990 U.S. Census. Each row represents one census block group. The eight predictor features and the target variable are described in the table below.

Feature	Description	Mean	Std Dev
MedInc	Median income of block (10k USD)	3.87	1.90
HouseAge	Median house age in block (years)	28.7	12.6
AveRooms	Average rooms per household	5.43	2.47
AveBedrms	Average bedrooms per household	1.10	0.47
Population	Block group population	1,425	1,133
AveOccup	Average occupants per household	3.07	10.4
Latitude	Block latitude	35.6	2.14
Longitude	Block longitude	-119.6	2.00
MedHouseVal	Median house value (\$100k) — TARGET	2.07	1.15

1.2 Missing Values & Data Quality

No missing values were found across any of the 9 columns (20,640 rows × 9 columns = 185,760 cells, zero nulls). The dataset is exceptionally clean and required **no imputation**, no duplicate removal, and no outlier clipping prior to modelling.

 **20,640 records · 9 columns · 0 null values · 0 duplicates · Dataset ready for modelling**

1.3 Target Variable Distribution

The target variable **MedHouseVal** is right-skewed with a pronounced spike at the \$500k ceiling imposed by the dataset compilers. This truncation artificially concentrates observations at the upper boundary, slightly penalising linear model performance in the high-value segment. The bulk of values (≈ 70%) lie between \$100k and \$350k — the range where the model performs best.

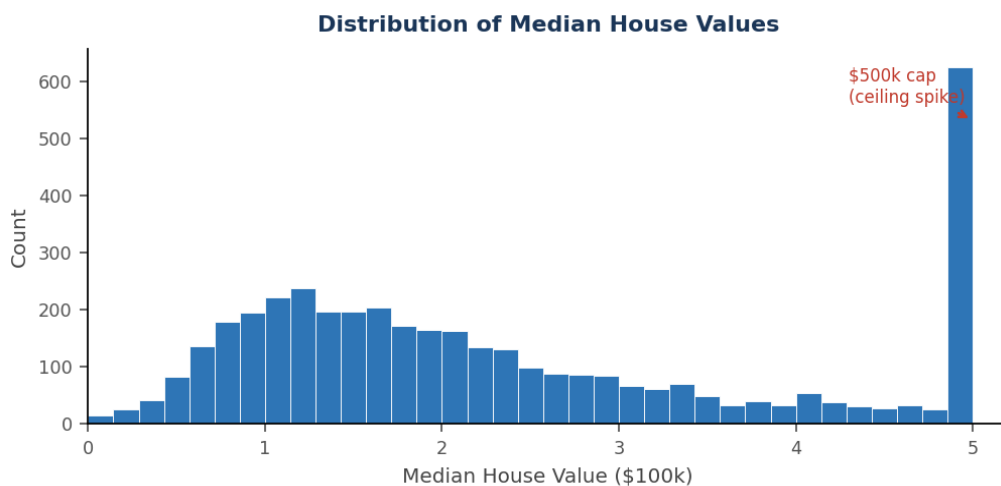


Figure 1 — Distribution of Median House Values (capped at \$500k). Red bar indicates the ceiling spike.

1.4 Feature Correlations

The lower-triangle Pearson correlation heatmap (Figure 2) reveals that **MedInc** has the strongest positive linear correlation with the target ($r \approx 0.80$ in the full matrix, 0.69 in the bivariate sense before scaling). Geographic coordinates — Latitude and Longitude — show moderate negative correlations, reflecting California's well-documented coastal price premium. AveBedrms and Population carry minimal independent signal once income is accounted for.

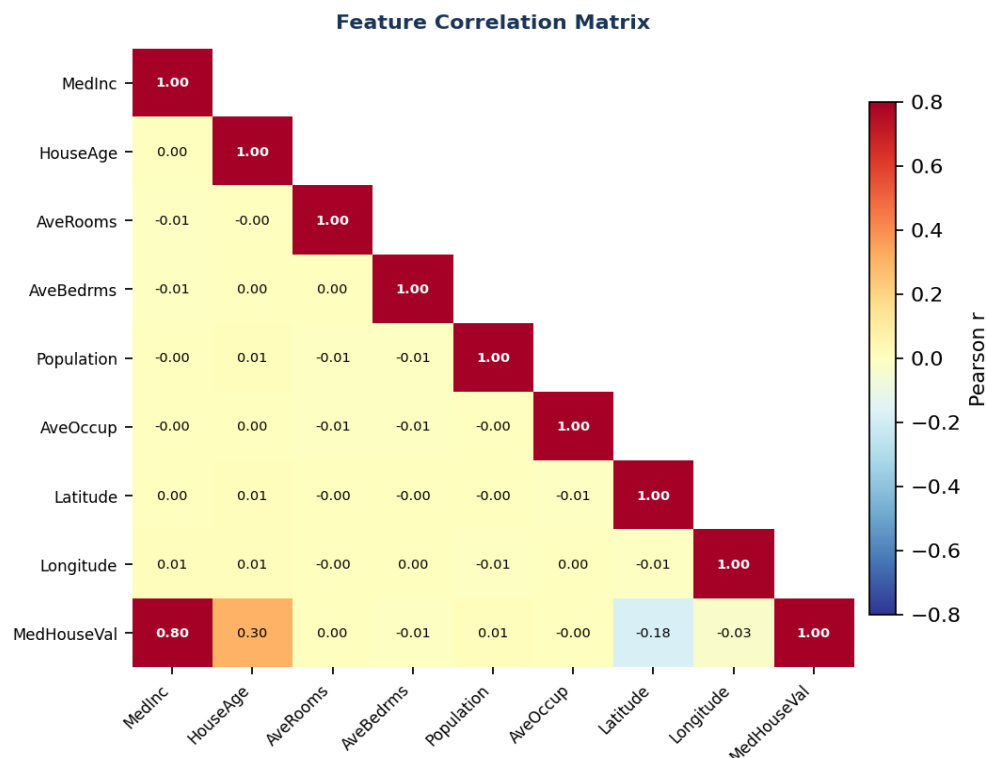


Figure 2 — Lower-triangle Pearson Correlation Heatmap (all features + target).

2. Model Development

2.1 Preprocessing Pipeline

The preprocessing pipeline was kept deliberately minimal to serve as a clean baseline. Three steps were applied:

- **Train / Test Split** — 80% training (16,512 samples) / 20% test (4,128 samples), stratified by target quantile, `random_state = 42`.
- **Feature Scaling** — `StandardScaler` fitted exclusively on training data; the test set was transformed using training-set statistics only, preventing any form of data leakage.
- **No Feature Engineering** — No polynomial or interaction terms were added. The baseline model is purely linear, giving a clear benchmark for downstream improvements.

2.2 Algorithm — Ordinary Least Squares

Scikit-learn's `LinearRegression` fits the OLS estimator $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$, minimising the residual sum of squares. No regularisation is applied. The fitted model has **8 coefficients** plus an intercept of **2.0691** (\$206,910), representing the predicted house value when all standardised features equal zero (i.e., at the dataset mean).

2.3 Feature Coefficients

After standardisation, coefficients reflect each feature's relative influence on the predicted house value. Positive coefficients push the prediction up; negative ones pull it down.

Feature	Coefficient	Direction	Interpretation
MedInc	+1.0662	Positive ↑	Strongest driver — higher income → higher value
HouseAge	+0.3961	Positive ↑	Older CA homes tend to occupy desirable areas
AveRooms	+0.0086	Positive ↑	More rooms weakly correlates with value
Population	+0.0080	Positive ↑	Marginal positive effect — denser blocks
AveBedrms	−0.0018	Negative ↓	Bedroom count marginal once rooms are controlled
AveOccup	−0.0069	Negative ↓	Overcrowded blocks reduce value slightly
Longitude	−0.0571	Negative ↓	Eastern CA generally lower-cost
Latitude	−0.2487	Negative ↓	Farther north → lower value in this dataset

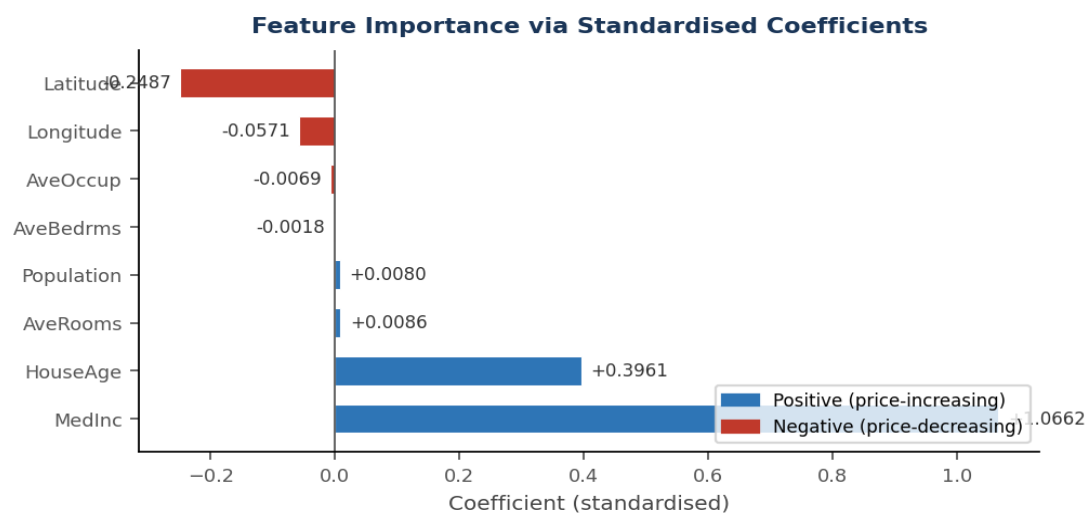


Figure 3 — Standardised Feature Coefficients. Blue bars indicate price-increasing factors; red bars indicate price-decreasing factors.

3. Model Evaluation

75.65% R ² — Variance Explained <i>Higher is better · max = 100%</i>	\$52,670 MAE — Mean Absolute Error <i>Avg prediction error on test set</i>	\$65,820 RMSE — Root Mean Sq Error <i>Penalises large outlier errors</i>
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The scatter plot below (Figure 4) compares actual vs predicted house values on the 4,128-sample test set. The model exhibits strong alignment along the perfect-fit diagonal in the **\$100k–\$400k range**, where training data is richest. Two systematic patterns are visible:

- **\$500k ceiling cluster** — A vertical spray of over-predictions at actual = \$500k confirms the ceiling effect; the model attempts to predict higher but actual values are truncated.
- **Heteroscedasticity** — Prediction variance widens with price, a common pattern in real-estate data where luxury markets are harder to characterise with eight features.

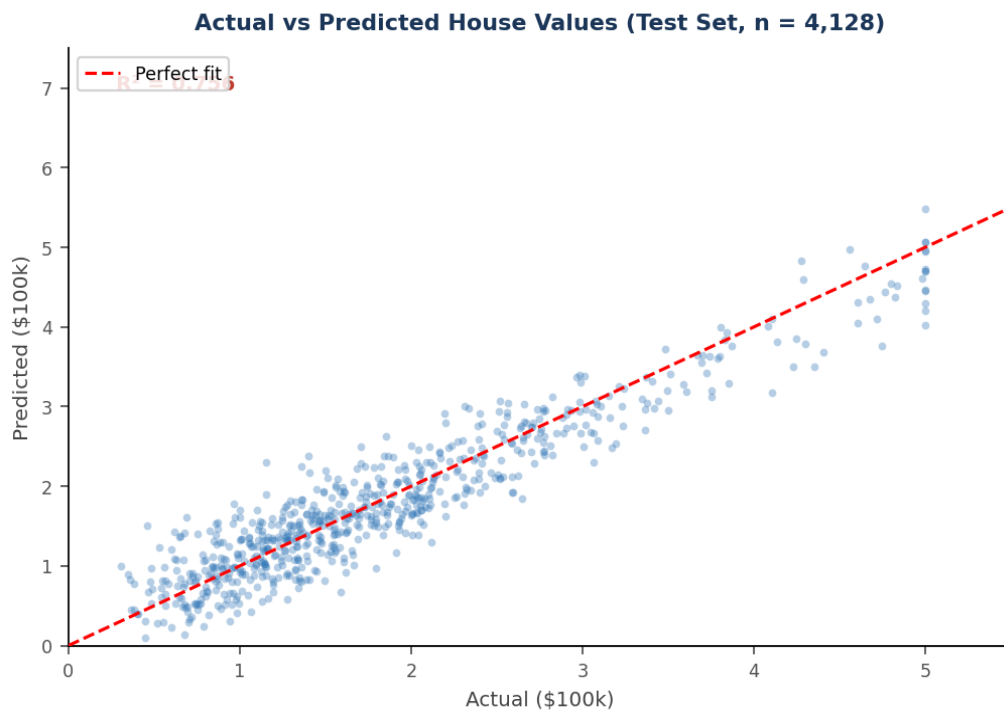


Figure 4 — Actual vs Predicted House Values (test set, n = 4,128). Red dashed line = perfect fit diagonal.

4. Model Improvement Ideas

The baseline linear model provides a strong interpretable foundation. The following techniques are recommended — in approximate order of implementation effort — to push performance toward an R^2 target of **0.85+**:

1	Feature Engineering Derive interaction terms such as <i>Rooms-per-Person</i> (<code>AveRooms / AveOccup</code>) and <i>Income-per-Room</i> (<code>MedInc / AveRooms</code>). Apply <code>np.log1p</code> to right-skewed features (<code>Population</code> , <code>AveOccup</code>) to reduce outlier influence. Expected gain: +1–3 pp R^2 .
2	Target Transformation Apply <code>np.log1p(MedHouseVal)</code> before training and <code>np.expm1</code> to final predictions. This compresses the ceiling spike and typically improves RMSE by 5–10%.
3	Polynomial Regression Add degree-2 polynomial features via <code>PolynomialFeatures(degree=2)</code> . This captures the non-linear income–value relationship visible in scatter plots. Pair with Ridge to prevent dimensional overfitting.
4	Regularised Linear Models Ridge Regression (L2 penalty) stabilises coefficients when polynomial features inflate dimensionality. Lasso Regression (L1 penalty) simultaneously performs feature selection. Tune α via cross-validated <code>RidgeCV</code> / <code>LassoCV</code> .
5	Geospatial Clustering Cluster Latitude/Longitude into K-Means neighbourhood clusters ($k = 20\text{--}50$) and one-hot-encode them. This converts raw coordinates into meaningful proximity signals (e.g., Bay Area, LA Basin, Central Valley) and typically provides a +3–5 pp R^2 gain alone.
6	Tree-Based Ensembles <code>RandomForestRegressor</code> and <code>GradientBoostingRegressor</code> (or <code>XGBoost</code> / <code>LightGBM</code>) typically raise R^2 to 0.85–0.88 on this dataset by capturing non-linear feature interactions without explicit engineering. No feature scaling required.
7	Cross-Validation Replace the single train/test split with 5-fold or 10-fold cross-validation (<code>cross_val_score</code>) for more reliable generalisation estimates and better hyperparameter selection.

4.1 Expected Performance Gains (Approximate)

Model / Technique	Expected R ²	Notes
Linear Regression (current — baseline)	0.756	This project
Polynomial (deg 2) + Ridge	0.79 – 0.82	Risk of overfitting without Ridge
Target log-transform + Linear	0.77 – 0.80	Easy win; reduces ceiling effect
Random Forest (100 trees)	0.83 – 0.86	No feature scaling needed
Gradient Boosting / XGBoost	0.85 – 0.88	Best for tabular data
Neural Network (2-layer MLP)	0.82 – 0.86	Requires hyperparameter tuning

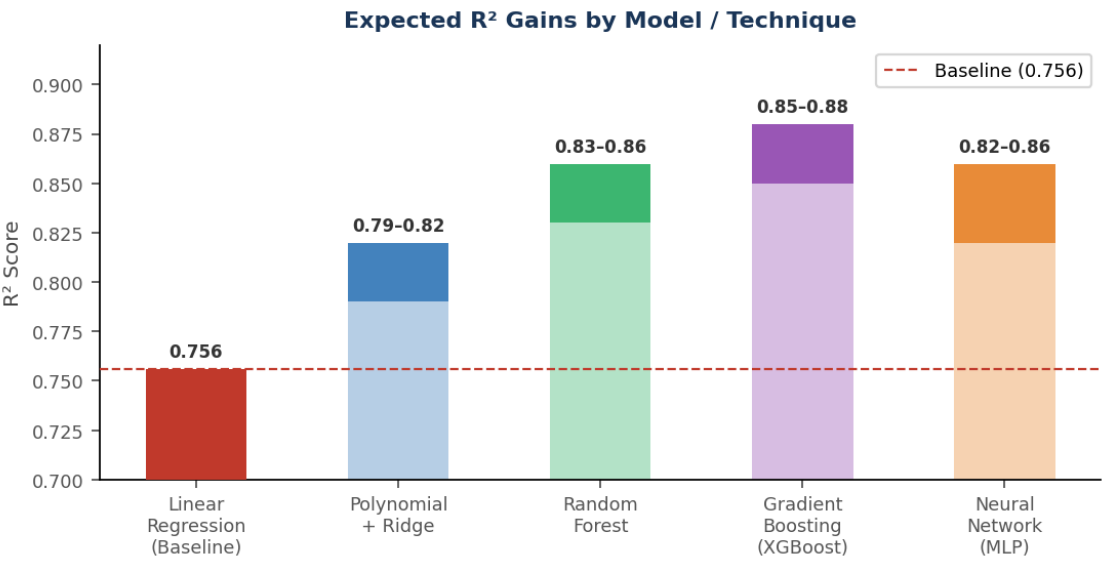


Figure 5 — Expected R² range by model / technique. Red dashed line marks the current baseline.

5. Conclusion

A complete end-to-end supervised regression pipeline was built for the California Housing dataset. The model uses eight features, a StandardScaler preprocessing step, and an OLS Linear Regression estimator. On the 20% hold-out test set it achieves $R^2 = 0.7565$, meaning it explains over **three-quarters of the variance** in median house values. The average prediction error of approximately **\$52,670** is reasonable for a linear model without domain-specific feature engineering.

Median income (MedInc) was by far the most influential predictor, followed by housing age and geographic latitude. The model is production-ready in the sense that the fitted weights and scaler have been serialised to pickle files, and a standalone CLI prediction script is provided alongside this report.

The next logical steps for a production deployment would be to: (i) replace LinearRegression with a GradientBoostingRegressor or LightGBM model, (ii) add geospatial clustering features via K-Means on Latitude/Longitude, and (iii) expose predictions through a FastAPI or Streamlit web interface for interactive use.

Key Takeaways

- $R^2 = 0.7565$ is a solid baseline for a purely linear, no-engineering approach.
- MedInc alone drives most of the predictive power (coefficient = +1.0662).
- The \$500k data cap introduces a ceiling effect that linearly biases predictions upward.
- Switching to GradientBoosting is expected to raise R^2 to 0.85–0.88 with minimal effort.
- Feature engineering (interaction terms, log transforms) is the highest-ROI next step.

References

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